

Bayesian Nonparametric Longitudinal Factor Model for Microbiome Counts

Shuangjie Zhang^{1*}, Seoyeon Ok², Robert S. Chapkin³, Yang Ni¹

¹Department of Statistics and Data Science, The University of Texas at Austin

²Department of Statistics, Texas A&M University

³Department of Nutrition, Texas A&M University

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Abstract

Longitudinal microbiome studies provide important insights into the temporal evolution of microbial abundances and dynamic cross-feature dependence. Modeling longitudinal microbiome data is challenging because microbiome counts are highly heterogeneous and zero-inflated, and, moreover, repeated measurements induce within-subject temporal dependence and possibly time-varying cross-feature covariance. To address these challenges, we develop a flexible Bayesian longitudinal factor model using a rounded log-normal latent representation. The mean abundance process begins with a Dirichlet process mixture baseline, which accommodates sparse and multimodal abundance distributions, and subsequently evolves through an autoregressive process. Autoregressive coefficients differ across prespecified biological groups, such as genotypes. A separate autoregressive factor process captures shared cross-feature dependence. For identifiability, an identity factor innovation covariance fixes the factor scale, while a lower-triangular loading matrix with positive diagonal entries identifies rotations, permutations, and signs. The model therefore distinguishes subject-feature mean evolution from shared dependence and yields a closed-form, time-indexed covariance. Simulation studies evaluate dynamic covariance recovery under matched and misspecified data-generating mechanisms, and an analysis of longitudinal mouse gut microbiome data illustrates the identifiable loading structure, distinct temporal processes, and dynamic covariance.

Keywords: Bayesian nonparametrics, Dynamic covariance, Longitudinal microbiome, Factor model, Dirichlet process mixture

*Address for Correspondence: 105 E 24th St, Austin, TX 78712. shuangjie.zhang@austin.utexas.edu

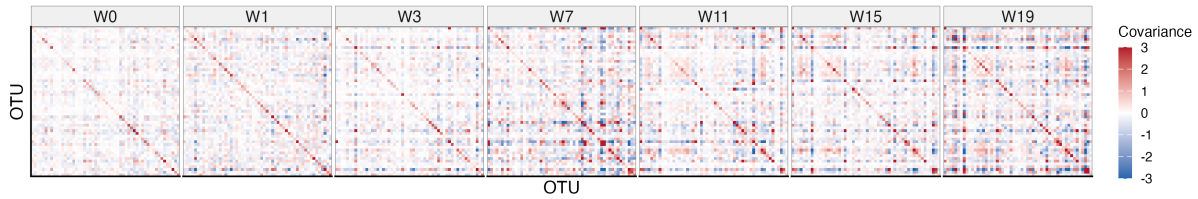
1 Introduction

High-throughput sequencing technologies in microbial ecology generate high-dimensional multivariate counts that characterize communities such as the gut, skin, and oral microbiome. The data are typically represented as a table of operational taxonomic units (OTUs) or amplicon sequence variants (ASVs) measured across biological samples (Li, 2015). Increasingly, these samples are collected repeatedly from the same subjects to study microbial development, resilience, disease progression, and responses to interventions (Gerber, 2015; Lyu et al., 2024; Creus-Martí et al., 2025). One important inferential objective is to determine how cross-feature dependence evolves over time.

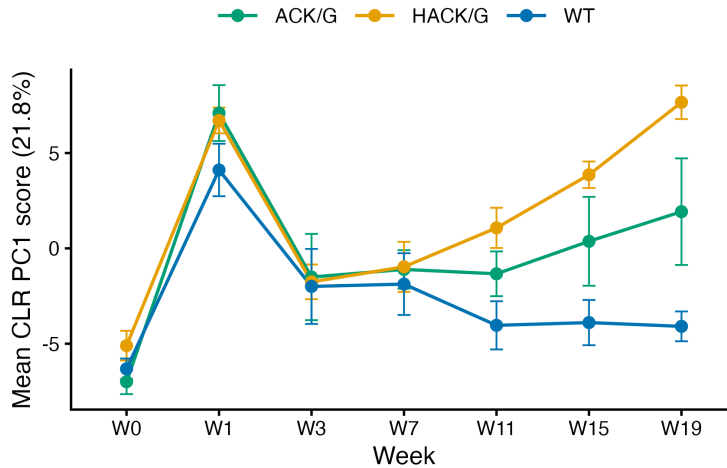
Generally, microbiome count data present several well-known statistical difficulties. Sequencing depth varies substantially across samples, excess zeros arise from both biological absence and limited detection, nonzero counts are highly overdispersed, and the number of taxa may be comparable to or larger than the number of subjects (Paulson et al., 2013; Li, 2015). Longitudinal sampling introduces two additional challenges. First, repeated samples from the same subject are temporally dependent, violating the independence assumption used by many cross-sectional procedures. Second, temporal variation contains at least two conceptually different components: a subject-feature mean trajectory describing how an individual taxon’s abundance changes, and a shared cross-feature process describing covariance variation among taxa. Without an explicit separation, feature-specific mean evolution can be absorbed into the estimated covariance and create misleading dependence patterns.

These challenges arise directly in our motivating longitudinal mouse gut microbiome study. The dataset contains 20 mice from three genotype groups, ACK/G (4), HACK/G (10), and WT (6), measured at weeks 0, 1, 3, 7, 11, 15, and 19 for 54 OTUs. Across the resulting 140 longitudinal samples, 18.9% of the observed counts are zero overall, with OTU-specific zero rates ranging from 0% to 92.1%. Figure 1(a) shows the empirical covariance matrix of the 54 CLR-transformed OTUs across the 20 mice at each time point.

Broad covariance patterns are shared across time points, while the magnitudes and signs of the covariances between specific OTU pairs vary visibly over time. Figure 1(b) presents genotype-specific mean scores and standard errors for the first principal component, obtained from a common PCA of all 140 CLR-transformed profiles. This component explains 21.8% of the total variation, and its trajectories show increasing separation among the genotype groups after week 7, particularly between HACK/G and WT. These patterns motivate a longitudinal model that allows feature interactions and count abundance to differ across known biological groups and evolve over time.



(a) Empirical OTU covariance at each time point.



(b) Genotype-specific CLR principal-component trajectories.

Figure 1: [Real Data] In (a), each heatmap shows the empirical covariance matrix among the 54 CLR-transformed OTUs, calculated across the 20 mice at the indicated time point. In (b), points and error bars represent genotype-specific means and standard errors of the first principal component scores obtained from a common PCA of all 140 CLR-transformed subject-time profiles.

Mixture models address some of the aforementioned challenges by representing samples or latent parameters as arising from multiple subpopulations. A prominent example

is the Dirichlet-multinomial mixture (DMM) (Holmes et al., 2012), which models each microbial composition using one of a finite number of Dirichlet-multinomial components. It accommodates unequal library sizes and overdispersion while clustering samples into latent metacommunity types. Gibbons et al. (2017) further extended DMM to a longitudinal subject-specific version applied to human gut microbiome time series data. Beyond finite-mixture formulations, Bayesian nonparametric priors have been used in microbiome analysis for several related but distinct purposes. Ren et al. (2017) constructed dependent normalized random measures with latent factors to jointly model microbial distributions and propagate uncertainty. Shuler et al. (2021) used dependent DP priors for covariate-indexed distributions of zero-inflation probabilities and differential-abundance effects in multivariate count regression. More recently, Sp-BGFM (Zhang et al., 2025) modeled multi-domain microbiome count vectors through a DP mixture of rounded multivariate log-normal kernels, coupling flexible baseline abundance distributions with a group factor structure for within- and cross-domain covariance. This formulation demonstrates how DP mixtures can accommodate zeros and substantial between-sample heterogeneity while avoiding an overly restrictive parametric baseline.

For modeling high-dimensional microbial dependence, existing methods largely fall into two categories. The first category estimates a sparse covariance or precision matrix after transforming the counts. SparCC (Friedman and Alm, 2012) estimates pairwise log ratio covariance between latent log-transformed abundances by approximating Pearson correlations. CCLasso (Fang et al., 2015) similarly uses log ratio transformed compositions but estimates a sparse and positive-definite log-basis covariance matrix through ℓ_1 -penalized least squares. SPIEC-EASI (Kurtz et al., 2015) applies a centered log-ratio transformation followed by sparse precision-matrix estimation, and Zi-LN (Prost et al., 2021) instead uses a CLR transformation and a latent Gaussian model to estimate the sparse precision matrix. See also REBACCA (Ban et al., 2015), COAT (Cao et al., 2019) and phyloBCG

(Chung et al., 2022). The second category assumes a latent low-dimensional representation of the high-dimensional covariance matrix. For example, ZI-MLN (Zhang et al., 2023) is a latent factor model that combines a rounded multivariate log-normal kernel with shrinkage priors on the loadings. Related low-rank and covariance methods have also been applied to transformed compositional and multi-omics profiles, including MOFA (Argelaguet et al., 2018), robust covariance estimation for compositional data (He et al., 2021), and zero-inflated logistic-normal principal components analysis (Zeng et al., 2023). These methods provide useful cross-sectional estimates, but for longitudinal microbiome data, fitting them independently at each time point ignores temporal dependence, whereas pooling all time points together ignores changes in covariance over time.

Methods developed specifically for longitudinal data have largely focused on modeling abundance trajectories. MC-TIMME uses negative binomial observation models with piecewise exponential mean dynamics and clusters OTUs by their responses through a Dirichlet-process mixture (Gerber et al., 2012). Multivariate sparse functional principal-component analysis instead represents subject-level multi-omics trajectories through smooth basis expansions (Jiang et al., 2022). These approaches characterize temporal changes in individual features but are not designed for time-varying covariance estimation. Related dynamic count models include hierarchical negative binomial state-space models (Chapados, 2014) and gamma-process dynamic Poisson factor analysis (Acharya et al., 2015). Although both approaches borrow information over time, their parametric count distributions are insufficiently flexible to accommodate the excessive zeros and heterogeneous abundance distributions common in microbiome data. MEFISTO introduces Gaussian-process factors for continuous multi-omics measurements (Velten et al., 2022), but analysis of zero-inflated microbial counts generally requires a transformation or a separate observation model. Thus, an important gap remains for a model that jointly provides a flexible count distribution, a distinct longitudinal mean process, and an identifiable factor process for time-varying

covariance.

We address this gap with LODI, a new Bayesian longitudinal factor model for microbiome counts that explicitly decomposes the temporal signal into a subject-feature mean process and a shared cross-feature factor process. We first use a rounded multivariate log-normal representation from [Canale and Dunson \(2011\)](#) to handle the discreteness of microbiome counts. Then, we let the initial value of the mean (of a temporal process) follow a Dirichlet-process mixture ([Sethuraman, 1994](#)), allowing heterogeneous and multimodal baseline abundances, and subsequently evolve through a group-specific AR(1) process. Next, we use a time-invariant loading matrix paired with an AR(1) factor process, resulting in a dynamic covariance. The abundance and factor components are distinguished through their different hierarchical and temporal structures. Beyond identified covariance estimation, LODI provides subject- and feature-specific abundance trajectories whose autocorrelation varies across groups (genotypes in our application). In parallel, the shared factor process yields within-time and lagged cross-time covariance matrices, allowing feature-pair dependence to be tracked over time.

The remainder of the paper is organized as follows. [Section 2](#) presents the model, prior specification, identifiability constraints, and posterior computation. [Section 3](#) evaluates LODI in simulation studies, and [Section 4](#) analyzes longitudinal mouse-gut microbiome data. [Section 5](#) concludes with directions for future work.

2 Model

2.1 Rounded log-normal observation model

Let Y_{ijt} denote the observed count for feature $j = 1, \dots, P$ from subject $i = 1, \dots, N$ at time $t = 1, \dots, T$. Each subject belongs to a known group $g(i) \in \{1, \dots, G\}$, such as a genotype. Following the rounded-kernel construction of [Canale and Dunson \(2011\)](#), we

introduce a log-normal positive latent variable

$$Y_{ijt}^* \sim \text{log-N}(\mu_{ijt}, \sigma_t^2), \quad (1)$$

where μ_{ijt} is the location parameter and σ_t^2 is the variance parameter. The observed count is obtained through $Y_{ijt} = \lfloor Y_{ijt}^* \rfloor$. Then, for every nonnegative integer y ,

$$\begin{aligned} \Pr\{Y_{ijt} = y \mid \mu_{ijt}, \sigma_t^2\} &= \int_y^{y+1} f_{\text{log-N}}\{y^* \mid \mu_{ijt}, \sigma_t^2\} dy^* \\ &= \Phi\{\log(y+1) \mid \mu_{ijt}, \sigma_t^2\} - \Phi\{\log y \mid \mu_{ijt}, \sigma_t^2\}. \end{aligned} \quad (2)$$

Here, $f_{\text{log-N}}$ denotes the log-normal density, $\Phi(\cdot \mid m, v)$ is the normal distribution function with mean m and variance v .

2.2 Longitudinal mean process

After specifying the rounded kernel, we decompose its location parameter as

$$\mu_{ijt} = s_{it} + \pi_{ijt} + \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_{it}. \quad (3)$$

The subject-specific term s_{it} is shared by all features measured from subject i at time t and absorbs multiplicative differences in sequencing depth or total sample abundance. In particular,

$$\mathbb{E}\{Y_{ijt}^* \mid s_{it}, \pi_{ijt}, \boldsymbol{\eta}_{it}\} = \exp\left\{s_{it} + \pi_{ijt} + \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_{it} + \frac{\sigma_t^2}{2}\right\}, \quad (4)$$

and hence $\exp\{s_{it}\}$ scales every feature in the same subject-time sample by the same amount. Thus, s_{it} is a latent sample-scale factor, which can be interpreted as the library size. From (4), the conditional likelihood alone does not identify s_{it} and π_{ijt} individually. For any scalar d , the transformation $s_{it} \mapsto s_{it} + d$, $\pi_{ijt} \mapsto \pi_{ijt} - d$, leaves μ_{ijt} unchanged. In the hierarchical model, we anchor this translation invariance by putting a prior with fixed

mean 0, which serves as a soft sum-to-zero constraint. The process π_{ijt} describes subject- and feature-specific abundance after normalization by s_{it} . Because it enters the model additively on the log scale, π_{ijt} corresponds to normalized baseline relative abundance. Equivalently, $\exp\{\pi_{ijt} - \pi_{iht}\}$ gives the baseline abundance ratio between features j and h . The final term in (3), $\boldsymbol{\lambda}_j^T \boldsymbol{\eta}_{it}$, captures lower-dimensional variation shared across features and is responsible for the non-diagonal covariance. We will discuss it in detail in Section 2.4.

2.3 Longitudinal Dirichlet-Process Mixture

Microbial abundances are often sparse, skewed, and multimodal across subjects and features. A common strategy for excess zeros is a zero-inflated model that combines a zero-generating mechanism with a parametric model for abundance. Examples include the zero-inflated Gaussian model for differential-abundance analysis (Paulson et al., 2013), the zero-inflated log-normal model for microbial association networks (Prost et al., 2021), and the Bayesian zero-inflated Dirichlet-multinomial regression model for multivariate compositional counts (Koslovsky, 2023). These formulations can distinguish structural absence from sampling variation, but they require a separate zero-generating mechanism and assumptions that identify it from the abundance component. Moreover, an observation attributed to the structural-zero component does not directly inform the positive-abundance density. In contrast, the rounded log-normal kernel in (2) assigns positive probability to zero through interval censoring. Consequently, zero and nonzero counts inform a common latent abundance distribution. To accommodate remaining multimodality without imposing a single parametric baseline distribution, we place a DP mixture (DPM) prior on the initial value of the log-normalized abundance. Specifically, using the stick-breaking

representation (Sethuraman, 1994), $w_\ell = z_\ell \prod_{h < \ell} (1 - z_h)$, $z_\ell \stackrel{iid}{\sim} \text{Be}(1, c)$, we have

$$\pi_{ij1} \sim \sum_{\ell=1}^{\infty} w_\ell \text{N}(\xi_\ell, \tau_\ell^2),$$

$$\pi_{ijt+1} = \phi_{g(i)} \pi_{ijt} + v_{ijt+1}, \quad v_{ijt+1} \sim \text{N}(0, V).$$

Here, $\phi_g \in (-1, 1)$ is the first-order autocorrelation parameter in AR(1) for group g , and V is the innovation variance. The group-specific AR(1) process propagates the DPM-distributed initial value π_{ij1} through time. We use the following conditionally conjugate priors $\xi_\ell \stackrel{iid}{\sim} \text{N}(\xi_0, \kappa_0^2)$, $\tau_\ell^2 \stackrel{iid}{\sim} \text{inv-Ga}(a_\tau, b_\tau)$, $\phi_g \stackrel{iid}{\sim} \text{N}(\mu_\phi, \sigma_\phi^2) \mathbb{1}\{-1 < \phi_g < 1\}$, $V \sim \text{inv-Ga}(a_V, b_V)$. Throughout, $\text{inv-Ga}(a, b)$ denotes an inverse-gamma distribution with shape a and rate b .

2.4 Longitudinal factor process

Recall from (4) that $\boldsymbol{\eta}_{it} \in \mathbb{R}^K$ is the factor process for subject i , and $\boldsymbol{\lambda}_j \in \mathbb{R}^K$ is the loading vector for feature j . Let $\boldsymbol{\Lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_P)^\top$. We assume that the factors follow an AR(1) process:

$$\boldsymbol{\eta}_{i1} \sim \text{N}(\mathbf{0}, I_K), \tag{5}$$

$$\boldsymbol{\eta}_{it+1} = \rho \boldsymbol{\eta}_{it} + \mathbf{u}_{it+1}, \quad \mathbf{u}_{it+1} \sim \text{N}(\mathbf{0}, I_K),$$

with autocorrelation parameter $\rho \sim \text{N}(\mu_\rho, \sigma_\rho^2) \mathbb{1}\{-1 < \rho < 1\}$. Since the likelihood function depends on the loadings and factors only through their product, and for any nonsingular $K \times K$ matrix H , $\boldsymbol{\Lambda} \boldsymbol{\eta}_{it} = (\boldsymbol{\Lambda} H) \{H^{-1} \boldsymbol{\eta}_{it}\}$, the loadings and factors are not separately identifiable without additional restrictions. Fixing both the initial and innovation covariance matrices of the factors at I_K fixes their scale and restricts this invariance to orthogonal transformations. To remove the remaining rotation, permutation, and sign ambiguities, we

constrain the first K rows of $\mathbf{\Lambda}$ to be lower triangular with positive diagonal:

$$\lambda_{jk} = 0 \quad \text{for } k > j, \quad \lambda_{jj} > 0 \quad \text{for } j = 1, \dots, K.$$

Provided that the leading $K \times K$ block has full rank, the triangular restriction fixes the factor orientation and ordering, while the positive diagonal fixes the signs (Geweke and Zhou, 1996; Lopes and West, 2004). We place independent $N(0, 1)$ priors on the free loading entries. Finally, we assume $\sigma_t^2 \stackrel{iid}{\sim} \text{inv-Ga}(a_\sigma, b_\sigma)$ for conjugacy.

Let $\tilde{\mathbf{Y}}_{it}^\star = \log(\mathbf{Y}_{it}^\star) = \{\tilde{Y}_{i1t}^\star, \dots, \tilde{Y}_{ijt}^\star, \dots, \tilde{Y}_{iPt}^\star\}^\top$. After integrating out the shared factor score, (5) implies that the within-time covariance of the latent log-abundance vector is

$$\mathbf{\Sigma}_t := \text{Cov}\left\{\tilde{\mathbf{Y}}_{it}^\star \mid s_{it}, \boldsymbol{\pi}_{it}\right\} = \frac{1 - \rho^{2t}}{1 - \rho^2} \mathbf{\Lambda} \mathbf{\Lambda}^\top + \sigma_t^2 I_P. \quad (6)$$

For a positive lag $\delta \geq 1$, the corresponding cross-time covariance is

$$\text{Cov}\left\{\tilde{\mathbf{Y}}_{it}^\star, \tilde{\mathbf{Y}}_{it+\delta}^\star \mid s_{it}, \boldsymbol{\pi}_{it}, s_{it+\delta}, \boldsymbol{\pi}_{it+\delta}\right\} = \rho^\delta \frac{1 - \rho^{2t}}{1 - \rho^2} \mathbf{\Lambda} \mathbf{\Lambda}^\top. \quad (7)$$

These latent covariance matrices also determine whether two observed counts are independent. In particular, conditional on s_{it} and $\boldsymbol{\pi}_{it}$, $\tilde{\mathbf{Y}}_{it}^\star$ is multivariate normal. Thus, for $j \neq j'$, $\Sigma_{jj',t} = 0$ implies that $\tilde{Y}_{ijt}^\star$ and $\tilde{Y}_{ij't}^\star$ are independent. Because the count transformation is applied separately to each coordinate, their transformed counts are also independent, and hence

$$\Sigma_{jj',t} = 0 \quad \implies \quad \text{Cov}\{Y_{ijt}, Y_{ij't} \mid s_{it}, \boldsymbol{\pi}_{it}\} = 0.$$

The same implication applies across time when the corresponding entry of (7) is zero. When a latent covariance is nonzero, the associated counts are generally dependent. We therefore conduct posterior covariance analyses on the latent log-abundance scale using (6) and (7).

Our formulation uses a common loading matrix $\mathbf{\Lambda}$ across time so that the same identified factor directions describe dependence throughout the longitudinal study. A more general specification could replace it by $\mathbf{\Lambda}_t$, giving within-time covariance $\{(1 - \rho^{2t})/(1 - \rho^2)\}\mathbf{\Lambda}_t\mathbf{\Lambda}_t^\top + \sigma_t^2 I_P$ and cross-time covariance $\rho^\delta \{(1 - \rho^{2t})/(1 - \rho^2)\}\mathbf{\Lambda}_t\mathbf{\Lambda}_{t+\delta}^\top$. Such a model permits pair-specific changes in covariance beyond the common temporal scaling in (6). However, estimating a separate $P \times K$ loading matrix at each time point is difficult when the number of subjects observed per time point is small relative to the number of features, as in microbiome studies with $N \ll P$. The time-invariant loading matrix therefore pools information across time and provides a more stable covariance estimate, whereas time-varying loadings would require additional temporal smoothing or shrinkage.

2.5 Posterior computation

Let Θ collect all unknown quantities. For an observed count y_{ijt} , we introduce an augmented latent Gaussian variable restricted to $[\log y_{ijt}, \log\{y_{ijt} + 1\})$, with lower endpoint $-\infty$ when $y_{ijt} = 0$. The augmented posterior is proportional to

$$\prod_{i=1}^N \prod_{j=1}^P \prod_{t=1}^T \mathcal{N}\left[\tilde{Y}_{ijt}^* \mid s_{it} + \pi_{ijt} + \boldsymbol{\lambda}_j^\top \boldsymbol{\eta}_{it}, \sigma_t^2\right] \times \mathbb{1}\left[\log y_{ijt} \leq \tilde{Y}_{ijt}^* < \log\{y_{ijt} + 1\}\right] p(\Theta). \quad (8)$$

Both longitudinal trajectories, $\{\pi_{ijt}\}_{t=1}^T$ and $\{\boldsymbol{\eta}_{it}\}_{t=1}^T$, are sampled jointly over time using forward-filtering backward-sampling (FFBS). One MCMC iteration proceeds as follows.

1. Sample each $\tilde{Y}_{ijt}^*$ from its normal full conditional truncated to $[\log y_{ijt}, \log\{y_{ijt} + 1\})$, using $\log(0) = -\infty$.

2. Sample the latent sample-scale term s_{it} . Under $s_{it} \sim N(m_s, v_s)$,

$$s_{it} \mid \cdot \sim N\{m_{s,it}, C_{s,t}\},$$

$$C_{s,t} = \left\{ \frac{P}{\sigma_t^2} + \frac{1}{v_s} \right\}^{-1}, m_{s,it} = C_{s,t} \left[\frac{m_s}{v_s} + \frac{1}{\sigma_t^2} \sum_{j=1}^P \left\{ \tilde{Y}_{ijt}^* - \pi_{ijt} - \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_{it} \right\} \right].$$

3. Sample the trajectory $\{\pi_{ijt}\}_{t=1}^T$ by FFBS, and update the mixture atoms, stick-breaking weights, group-specific coefficients ϕ_g , and innovation variance V .
4. Sample the free elements of $\mathbf{\Lambda}$ subject to its triangular and positive-diagonal restrictions. Update each trajectory $\{\boldsymbol{\eta}_{it}\}_{t=1}^T$ by FFBS, center the factor scores across subjects at every time point, and then update ρ .
5. Independently for each time point, sample

$$\sigma_t^2 \mid \cdot \sim \text{inv-Ga} \left[a_\sigma + \frac{NP}{2}, b_\sigma + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^P \left\{ \tilde{Y}_{ijt}^* - s_{it} - \pi_{ijt} - \boldsymbol{\lambda}_j^T \boldsymbol{\eta}_{it} \right\}^2 \right].$$

Complete full-conditional derivations are provided in Supplementary Materials Section A. The sampler is implemented in Rcpp and RcppArmadillo for computational efficiency. An R package is available on <https://shuang-jie.github.io/LODI>.

3 Simulation

3.1 Setup and evaluation

We evaluate LODI's performance with simulations. We first test whether LODI identifies factors and recovers the true covariance under matched-model and ZINB-misspecified conditions. Then, we utilize real-data features in Section 4 to generate a simulation that mimics real, challenging data. In all simulations, we generate longitudinal count data $\mathbf{Y} \in \mathbb{Z}_{\geq 0}^{N \times T \times P}$ and measure recovery of the covariance $\boldsymbol{\Sigma}_t$ implied by the truth using

$\text{RMSE}_t = \sqrt{\sum_{j,j'} (\hat{\Sigma}_{jj',t} - \Sigma_{jj',t}^{\text{true}})^2 / P^2}$. We report RMSE averaged across time within each replicate and then across the independent replicates available for each simulation setting.

We compare LODI to three existing covariance estimators for microbiome data: (i) SP-BGFM (per-time) (Zhang et al., 2025), a Sparse Bayesian Group Factor Model fit separately at each timepoint; (ii) SP-BGFM (joint-fit), the same model with a joint fit on the covariance without temporal dependence on the mean; and (iii) MEFISTO (Velten et al., 2022), a factor model with a Gaussian process on longitudinal covariates. SP-BGFM is fitted directly to the observed counts, whereas MEFISTO uses $\log(Y + 0.5)$ as Gaussian input.

The implementation fixes $(\xi_0, \kappa_0^2, a_\tau, b_\tau, \mu_\phi, \sigma_\phi^2, a_V, b_V, c) = (0, 25, 2, 1, 0, 1, 2, 1, 1)$. Thus, the mixture-location prior has standard deviation 5, and the DP concentration parameter is fixed at $c = 1$. We test the sensitivity of these parameters in Supplementary Materials Section C. The results are not sensitive to the hyperparametric choices. The implemented sampler uses a finite truncation with L mixture components, and the simulation and data analyses use $L = 30$. A component diagnostic assessing the adequacy of this truncation level for the real-data fit is provided in Supplementary Materials Section D. We also examine the sensitivity of LODI to the number of factors $K \in \{4, 6, 8, 10\}$ when the data-generating truth has $K^{\text{tr}} = 4$. The results below are under $K = K^{\text{tr}}$, and results under other K 's are presented in Supplementary Materials Section C. The estimates of factor loadings and covariance RMSEs are also not sensitive to the choice of K neither. Other representative MCMC trace plots and additional parameters and covariance results are provided in Supplementary Materials Section B.

3.2 Simulation results

Simulation 1 generates counts directly from the LODI model to evaluate mean-process and covariance recovery under matched conditions and to assess sensitivity to the fitted number

of factors. The data-generating design uses $N = 20$, $T = 7$, $P = 50$, $K^{\text{tr}} = 4$, and $G = 3$. The baseline π_{ij1} has three mixture components with locations $\{-5, 3, 8\}$, common variance 0.5, and weights $\{0.4, 0.3, 0.3\}$. We set the true abundance-process AR(1) coefficients at $\phi_1 = -0.8$, $\phi_2 = 0.9$, and $\phi_3 = 0.7$ for the three groups, respectively, with innovation variance $V^{\text{tr}} = 0.2$. These values represent negative temporal dependence in the first group and positive temporal dependence in the other two groups. The factor process has $\rho^{\text{tr}} = 0.7$, initial covariance I_K , and innovation covariance I_K . The log-scale variance $\sigma_t^{2,\text{tr}}$ is drawn independently from $\text{Unif}(0, 1)$. Finally, counts are generated as $y_{ijt} = \lfloor \exp\{\tilde{Y}_{ijt}^*\} \rfloor$. We fit LODI using $K \in \{4, 6, 8, 10\}$ and fit each comparator using its default specification with $K = K^{\text{tr}}$.

Figure 2 illustrates recovery of the mean process and the loadings. In (a), the posterior mean of $\mu_{ijt} = s_{it} + \pi_{ijt}$ closely follows its true value across subjects, features, and times. The group-specific AR coefficients are also accurately recovered, with posterior means (SDs) of -0.801 (0.004), 0.910 (0.004), and 0.694 (0.006), compared with the true values -0.8 , 0.9 , and 0.7 , respectively. A good estimate of the mean provides a foundation for the covariance estimate. We compare entries of $\mathbf{\Lambda}$ directly with the truth in (b) since both the true and fitted loading matrices obey the same lower-triangular, positive-diagonal constraint. No post-hoc rotation or sign alignment is used. This leads to a good estimate of covariance in Table 1.

LODI substantially outperforms all three comparators and remains stable under moderate over-specification of the factor dimension, as shown in Table 1. The estimates of covariance for $K = 4, 6, 8$, and 10 are nearly indistinguishable relative to their average RMSE. The details of other parameter estimates and different K are provided in Supplementary Materials Section C. These results indicate that covariance is identifiable and not sensitive to over-specification of K .

In Simulation 2, we generate data from a zero-inflated negative binomial distribution.

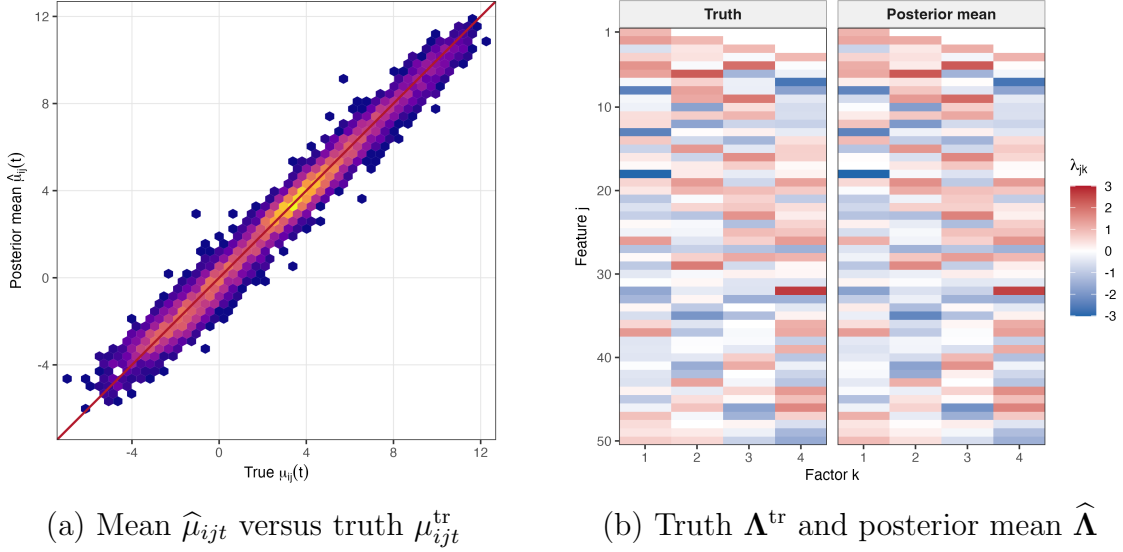


Figure 2: [Simulation 1] (a) Posterior mean versus truth for all $N \times T \times P = 7,000$ values of the longitudinal mean process μ_{ijt} ; warmer color represents a larger number of overlapping values, and the red line is the identity line. (b) Heatmaps of the true and posterior mean loading matrices with features in rows and factors in columns.

Method	Fitted K	Average RMSE (SE)
LODI	4 (truth)	0.840 (0.024)
LODI	6	0.853 (0.020)
LODI	8	0.837 (0.020)
LODI	10	0.883 (0.021)
MEFISTO	4	3.165 (0.036)
SP-BGFM (per-time)	4	3.557 (0.042)
SP-BGFM (multi-domain)	4	3.742 (0.046)

Table 1: [Simulation 1] RMSE of Σ_t covariance recovery is averaged across time within each replicate and then across 50 independent replicates. Parentheses report its standard error.

Let $z_{ijt} \sim \text{Bern}(\pi)$ denote structural absence. If $z_{ijt} = 1$, then $y_{ijt} = 0$; otherwise $y_{ijt} \sim \text{NB}(\vartheta_{ijt}, \phi_{\text{NB}} = 3)$. Since the negative binomial distribution does not have a covariance parameter, we put the latent factors on the log of the latent mean. $\log \vartheta_{ijt} = c_i + \boldsymbol{\lambda}_j^{\text{T}} \boldsymbol{\eta}_{it}$ and $c_i \sim \text{N}(3, 1)$. The factor process has $\rho = 0.7$ and identity innovation covariance. Thus the data-generating likelihood is misspecified relative to LODI. We use $N = 20$, $T = 5$, $P = 50$, $K^{\text{tr}} = 4$, $G = 3$, and test the zero inflation rate $\pi \in \{0.10, 0.30, 0.50\}$.

Method	$\pi = 0.10$	$\pi = 0.30$	$\pi = 0.50$
LODI	0.954 (0.029)	1.205 (0.036)	1.657 (0.053)
MEFISTO	2.194 (0.068)	2.672 (0.067)	2.974 (0.085)
SP-BGFM (per-time)	2.241 (0.073)	2.849 (0.083)	3.290 (0.085)
SP-BGFM (multi-domain)	2.477 (0.082)	2.926 (0.072)	3.298 (0.092)

Table 2: [Simulation 2] RMSE of Σ_t covariance recovery under increasing zero inflation is averaged across time within each replicate and then across 50 replicates. Parentheses report its standard error.

As reported in Table 2, LODI still performs best across the three values of zero rates π even under a misspecified model. This advantage over SP-BGFM cannot be attributed to count preprocessing, because both methods use rounded-log-normal augmentation for the raw counts. Instead, the comparison highlights LODI’s autoregressive factor process, which allows the sharing of information across time for estimating the evolution of Σ_t . Fitting SP-BGFM separately at each timepoint discards the information sharing, whereas treating timepoints as multiple domains does not impose an ordered temporal process. MEFISTO models smooth temporal factors but is fitted to $\log(Y + 0.5)$ without a flexible mean process. Additional results for Simulation 2 are provided in Supplementary Materials Section B.

Simulation 3 evaluates a real-data scale simulation under the zero-inflated negative binomial misspecification introduced in Simulation 2. In real data Section 4, we have 20 samples collected on 54 OTUs from 3 groups and 7 time points. Thus, we use $N = 20$, $T = 7$, $P = 54$, $K^{\text{tr}} = 8$, and $G = 3$, and generate the latent log mean as $\log \vartheta_{ijt} = c_i + \lambda_j^{\text{T}} \eta_{it}$. Motivated by the structured empirical OTU dependence shown in Figure 1, we set $K^{\text{tr}} = 8$ to generate a moderately complex low-rank covariance structure. To reproduce the count scale, we draw $c_i \sim N(\bar{\ell}, s_{\ell}^2)$ using the empirical log-library-size mean $\bar{\ell} \approx 10.76$ and empirical standard deviation $s_{\ell} \approx 0.77$ obtained from the real data. We

set the structural-zero probability to the empirical zero rate, $\pi = 0.189$, and use a factor process with $\rho = 0.7$ and identity innovation covariance. We let the negative binomial dispersion parameter equal to $\phi_{\text{NB}} = 0.745$, obtained by averaging the valid feature specific method-of-moments estimates from the positive real-data counts.

Figure 3 examines temporal mean trajectories and posterior predictive check for goodness of fit. Panel (a) shows four posterior mean estimates of mean trajectories (solid), tracking closely the true normalized latent log-mean processes (dashed) with distinct temporal shapes. For the posterior predictive check, we compare summaries of posterior predictive samples with the corresponding summaries of the observed or simulated counts. Panels (b)–(d) provide distributional checks across increasing sparsity. Panel (b) evaluates the positive-count distribution in a setting with no observed zeros, and Panels (c) and (d) represent increasingly sparse settings. As the zero rate in the observed data increases, the posterior predictive distribution adaptively places more mass near zero. This pattern indicates that the rounded kernel reproduces zero-rich count behavior without a separate zero-inflation indicator.

Table 3 evaluates covariance recovery for LODI and comparators across all replicates. LODI has the smallest average RMSE. The absolute RMSE values are not directly comparable with those in Simulation 2 because Simulation 3 uses different dimensions and a distinct real-data-calibrated data-generating setting.

Method	Average RMSE (SE)
LODI	2.002 (0.039)
MEFISTO	2.376 (0.057)
SP-BGFM (per-time)	2.284 (0.074)
SP-BGFM (multi-domain)	2.976 (0.083)

Table 3: [Simulation 3] RMSE of Σ_t covariance recovery is averaged across time within each replicate and then across 50 replicates, and parentheses report standard errors.

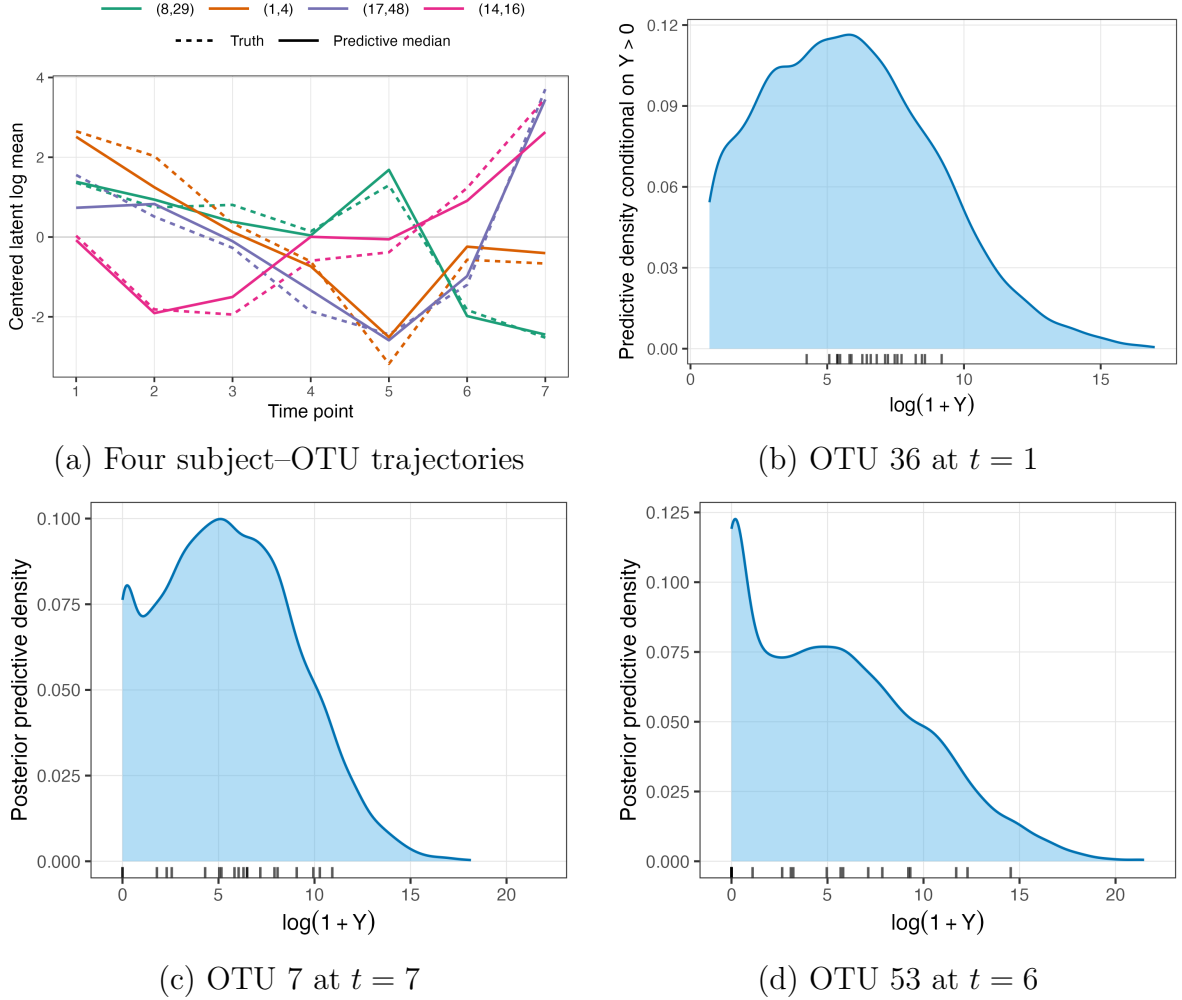


Figure 3: [Simulation 3] Panel (a) compares true normalized latent log-means (dashed) with posterior mean estimates (solid) for $(i, j) = (8, 29)$, $(1, 4)$, $(17, 48)$, and $(14, 16)$. Panels (b)–(d) show posterior predictive densities of $\log(1 + Y_{ijt}^{\text{rep}})$ for OTU 36 at $t = 1$, OTU 7 at $t = 7$, and OTU 53 at $t = 6$, respectively. Black rug marks denote the 20 observed counts.

4 Real data analysis

We apply LODI to a longitudinal mouse-gut microbiome dataset comprising $N = 20$ mice measured at $T = 7$ timepoints across $P = 54$ OTUs. The mice belong to three genotype groups: ACK/G (4), HACK/G (10), and WT (6). The raw count matrix has a total observation count summing to $\approx 1.6 \times 10^7$, with library sizes per (subject, time) ranging from 1.0×10^4 to 3.1×10^5 . The overall zero rate is 18.9%, but per-feature zero rates are highly heterogeneous (0% to 92.1%). We fit LODI with $K = 10$ latent factors and 400,000 MCMC iterations.

To assess goodness of fit, we conduct posterior predictive checks and compare the fitted abundance process to the observed one. Figure 4 shows the posterior predictive density of $\log(Y_{ij}^{\text{rep}} + 1)$ for six representative OTUs ordered by their observed zero rates. LODI’s posterior predictive distribution flexibly captures the shape of the count distribution for abundant OTUs with few zeros (e.g., OTU 32 at Week 7, zero rate 0%), bimodal OTUs (OTU 52 at Week 3, zero rate 18%), and sparse OTUs (e.g., OTU 39 at Week 1, zero rate 55%). This flexibility benefits directly from LODI’s DP-mixture prior on π_{ij1} .

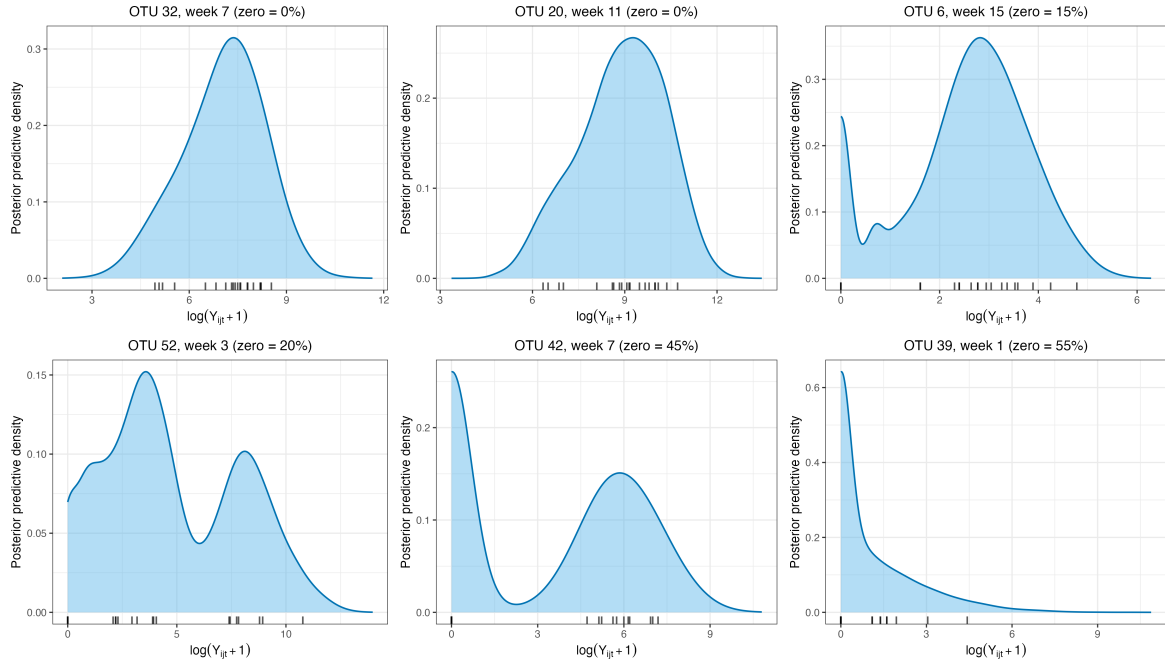


Figure 4: [Real Data] Posterior predictive densities (blue) of $\log(Y_{ijt}^{\text{rep}} + 1)$ for six representative OTU–time pairs, ordered by increasing time-specific observed zero rate. Each panel selects one OTU and one week and pools predictions across the 20 mice; black rug marks show the corresponding 20 observed counts. OTU 52 at week 3 illustrates multimodal flexibility, OTU 6 at week 15 represents moderate sparsity, and OTUs 42 and 39 illustrate recovery under substantial sparsity.

Figure 5(a) illustrates the fitted normalized abundance process for four subject–OTU pairs. The empirical curves are smoothed log-relative abundances, and the fitted curves are posterior means of the normalized latent process $\pi_{ijt} + \lambda_j \eta_{it}$. Comparison of raw π_{ijt} is included in Supplementary Materials Section D. Both are centered over time to compare their temporal shapes. The posterior trajectories reproduce several rises, declines, and reversals

in the observed data. These posterior checks indicate that LODI distinguishes the two underlying mechanisms well. The posterior means of ϕ_g , which capture the feature-specific abundance process π_{ijt} , are 0.993, 0.994, and 0.995 across the three genotype groups, and the innovation variance is $V = 0.473 (\pm 0.040)$. The estimated innovation variance allows departures from the preceding abundance level. The shared factor process has posterior mean $\rho = 0.347 (\pm 0.065)$ and 95% credible interval $[0.221, 0.474]$, indicating greater temporal variation in the shared cross-OTU dependence structure. Figure 5(b) displays the posterior mean of the identified loading matrix. Its effective rank is substantially smaller than the prefixed $K = 10$. Measured by $\|\hat{\boldsymbol{\Lambda}}_k\|^2 / \|\hat{\boldsymbol{\Lambda}}\|_F^2$, factors 1, 3, and 5 contribute 56.4%, 21.6%, and 8.2%, respectively, whereas each remaining factor contributes only 1.2%–2.7% of the total magnitude.

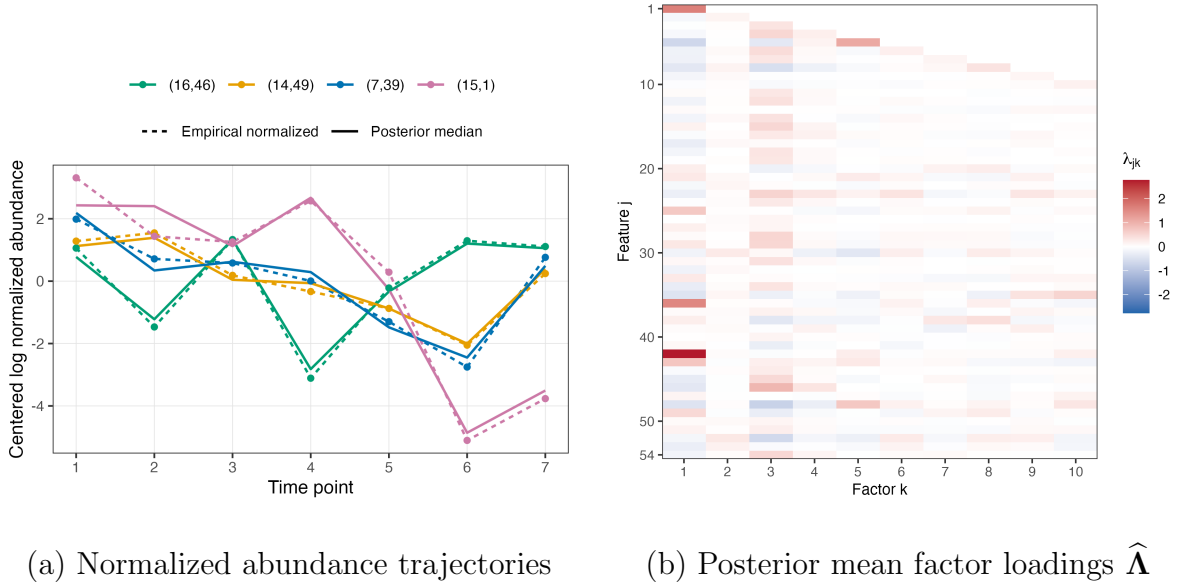


Figure 5: [Real Data] In (a), empirical smoothed log-relative abundances (dashed, with points) are compared with posterior mean normalized latent trajectories (solid) for four illustrative subject–OTU pairs; each trajectory is centered over time. Panel (b) shows the posterior mean of the identified loading matrix $\hat{\boldsymbol{\Lambda}}$.

Motivated by the three dominant loading columns in Figure 5(b), Figure 6 shows posterior mean trajectories of η_{ikt} for factors 1, 3, and 5. Thin curves represent individual subjects, and thick curves represent genotype-group averages. Factor 1 exhibits the strongest

short-term reversals, including a decline from W1 to W3 followed by recovery at W7 for the ACK/G group and a decline at W7 followed by recovery at W19 for the WT group. Factor 3 shows increasing separation after W7: the WT group moves upward, reaching 0.71 at W19, whereas HACK/G moves downward to -0.46 . Factor 5 remains closer to zero at the group level, although individual subjects still fluctuate.

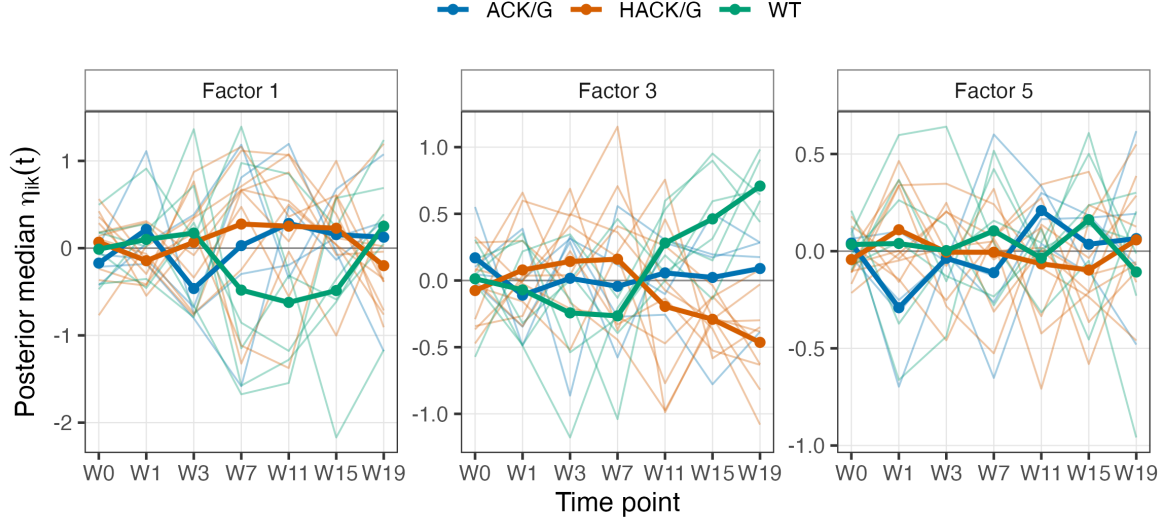


Figure 6: [Real Data] Posterior mean factor-score trajectories η_{ikt} for the three dominant loading directions. Thin curves show individual subjects and thick curves show genotype-group averages. Factor scores are centered across subjects at each time point.

The magnitude of λ_{jk} measures how strongly OTU j participates in factor k , while its sign determines whether the OTU changes in the same or opposite direction as the factor score. Factor 1 is driven most strongly by OTU 42 (family Clostridiaceae; posterior mean loading 2.77), OTU 1 (genus *Clostridium*; 1.65), and OTU 36 (order RF39; 1.56). The largest absolute loadings for factor 3 occur for OTU 46 (genus *Butyricicoccus*; 0.93), OTU 48 (genus *Enterococcus*; -0.77), and OTU 52 (genus *Akkermansia*; -0.64). Factor 5 is driven primarily by OTU 5 (family Enterobacteriaceae; 1.12), followed by OTU 48 (genus *Enterococcus*; 0.71) and OTU 30 (genus *Blautia*; -0.40).

We also report the posterior correlation estimates over time and the strongly associated OTU pairs in Supplementary Materials Section D. Several estimated associations are consis-

tent with previous gut-microbiome studies. In particular, the positive association between OTU 20 (genus *Allobaculum*) and OTU 49 (genus *Bifidobacterium*) ranges from 0.303 to 0.780 across weeks. This agrees with the significant positive co-abundance relationship reported by Wang et al. (2019) across 101 healthy mice and with the Spearman correlation of 0.78 reported by Tachon et al. (2013) in aged mice. The positive association between OTU 20 and OTU 38 (family Coriobacteriaceae), with posterior mean correlations ranging from 0.318 to 0.706, is also consistent at a broader taxonomic level with the positive association between Erysipelotrichaceae—the family containing *Allobaculum*—and Coriobacteriaceae reported by Liu et al. (2022). Finally, the consistently positive association between OTU 5 (family Enterobacteriaceae) and OTU 48 (genus *Enterococcus*), with posterior mean correlations ranging from 0.341 to 0.483, is concordant with the identification of *Enterococcus* species as prominent co-colonizers of Enterobacteriaceae in a large-scale analysis of human gut metagenomes (Yin et al., 2025). These agreements support the biological plausibility of the estimated dependence structure, although the correlations do not necessarily represent direct microbial interactions and may partly reflect shared host or environmental effects.

5 Discussion

We have introduced LODI, a Bayesian longitudinal factor model for microbiome counts. It combines a rounded log-normal latent representation of count data, a DP-mixture baseline, a group-specific AR(1) abundance process π_{ijt} , and a shared AR(1) factor process η_{it} . LODI is capable of capturing both the temporal evolution of microbial abundances and dynamic cross-feature dependence. Posterior inference uses FFBS updates for both longitudinal processes within an Rcpp-based Gibbs sampler. Simulations under both the assumed model and misspecified zero-inflated negative binomial mechanisms demonstrate improved covariance recovery over existing methods, and the mouse-gut microbiome analysis illustrates how LODI separates persistent subject-feature abundance trajectories from

shared variation across OTUs.

Several extensions could further increase flexibility while preserving this separation. The current model uses a time-invariant loading matrix $\mathbf{\Lambda}$, which pools information across time and makes the factor directions directly comparable. A dynamic loading process $\mathbf{\Lambda}_t$ could permit the OTU-specific dependence pattern itself to evolve, rather than allowing temporal covariance changes only through the factor process and residual variance. Because an unrestricted $P \times K$ loading matrix at every time point is hard to estimate when $N \ll P$, a practical version would require temporal smoothing, shrinkage toward a common loading matrix, and identifying constraints that remain coherent over time.

The baseline distribution could also vary across biological groups or covariates. A hierarchical Dirichlet process could define genotype- or OTU-specific mixing distributions that share atoms but use different weights (Teh et al., 2006). Alternatively, a dependent Dirichlet process could allow the entire baseline distribution to change flexibly with genotype or other subject-level covariates (Quintana et al., 2022). These extensions may reveal group-specific baseline heterogeneity, although reliable uncertainty quantification would require more subjects within each group.

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